

Fail-Closed Cloud Training of a HalfKAv2_hm NNUE under Live Persona Adaptation: Speed Choices that Do Not Spend Elo

Unchessed AI Technical Report TR-2026-08-31b | IEEE-style preprint | 31 August 2026

Abstract?

We specify the exact A100 training stack Unchessed will run in the cloud: GPU-resident shards, batch 131072, TF32, bfloat16 autocast, fused Adam, cuDNN benchmark, early-stop on val-MAE, best-checkpoint export, and a fail-closed preflight that refuses to start unless Adaptive/persona stays on, UnarchitectedHint stays off, and the operator types `GO_CLOUD=I_ACCEPT_SPRT_GATES`. Speed knobs do not change the WDL 2.5 loss or the feature scheme. Quality is defined as SPRT, not as a small printed val-MAE. Pipeline certainty is 100% for those gates; Elo is not.

Index Terms? NNUE, A100, TF32, mixed precision, SPRT, UCI Adaptive, cloud preflight.

I. INTRODUCTION

Billing on an A100 is wall-clock. The previous cloud launcher still labelled the run v3, could export a last-checkpoint net, and could start without a go token. This revision makes the launcher v4, fail-closed, persona-on, and as fast as Ampere allows without touching the loss. We do not claim 100% Elo. We claim 100% pipeline certainty: a billed run cannot start unless preflight passes, and Adaptive cannot be turned off by that run.

II. NON-NEGOTIABLE QUALITY

Loss remains $|\text{sigmoid}(\text{raw}) - t|^2.5$ with $t = 0.7$ $\text{sigmoid}(\text{cp}/400) + 0.3$ ($\text{wdl}/2$), the Stockfish nnue-pytorch recipe already used here. Features remain HalfKAv2_hm with factorization; export is UNCHNNUE v4 (22528 x 256 + 8-bucket head). The epoch argument is a cap of 15; early-stop patience 3 and min-delta 0.1 cp export the best checkpoint, which is the round-13 fix for three SPRTs that had shipped a worse last epoch. Adaptive stays default true. UnarchitectedHint stays default false. A net is promoted only after SPRT versus unchessed-nnue.bin at $\text{tc}=10+0.1$ with Adaptive on. That is the quality bar. A small val-MAE is not.

III. SPEED CHOICES (WHY THESE, NOT OTHERS)

GPU-resident data. One host-to-device copy of the 104-byte records; shuffle is randperm on device. This removes the PCIe round-trip that made CPU training the slow path on the WSL box. Batch 131072 is the production A100 size already in full_pipeline_cloud.sh. The net is tiny relative to 80 GB; occupancy of EmbeddingBag is the limit, so larger batches are the first-order speed lever. TF32 tensor cores are the Ampere default for matmul; EmbeddingBag integer indices are unchanged and exported weights stay IEEE-754 f32. bf16 autocast covers forward and loss; Adam keeps fp32 master weights. No GradScaler: bf16 shares fp32's exponent. Indices are int64 and sit outside autocast. Opt-out with `USE_AMP=0` if a future kernel miscasts. Fused Adam is one CUDA kernel for the moment update, with TypeError fallback. cuDNN benchmark is kept for the Linear head. torch.compile is OFF by default: EmbeddingBag plus dynamic nnz has compiled poorly in the field. `zero_grad(set_to_none=True)` avoids writing zeros across 11M+ embedding rows every step. Early-stop is the cheapest quality win: the 100-epoch v3 rerun paid for 35 useless epochs after the 53.6 cp minimum. None of these knobs change the loss.

IV. FAIL-CLOSED PREFLIGHT

tools/nnue_cloud_runtime.py returns a blocking error list if PERSONA_ACTIVE is not 1, if UNARCH_HINT is 1, if record count is below 1000 or above 500 million, or if the go token is required and missing. scripts/nnue-pipeline/cloud_train_v4.sh exports `REQUIRE_CLOUD_GO=1`. An empty `GO_CLOUD` in the monthly pipeline now fails training instead of silently starting a v3 run. That is the 100% certainty this paper is allowed to claim: the billed process cannot begin in a persona-off, hint-on, or un-acknowledged configuration.

V. PERSONA STAYS ACTIVE

MATCH / PUNISH / CLINCH / DEFEND live in adapt.rs and read search eval_cp, not trainer val-MAE. Training a stronger net with Adaptive left at its UCI default is the configuration the SPRT must use. Turning Adaptive off for a cleaner Elo number is a different experiment and is not this launcher. Simulation in the companion report showed mode-flip rate falling from 30.5% at 80 cp eval MAE to 4.5% at 10 cp: a more accurate evaluator stabilises persona, it does not disable it.

VI. WHAT THIS DOES NOT CLAIM

Sub-20 cp val-MAE on 5000-node HCE labels remains unreachable (Gaussian Bayes floor about 56 cp; repo best 51.1 cp at 27M unique). Speed knobs do not lower that floor. Cloud 178M is still gated on a local 108M best-checkpoint SPRT for Elo. This paper only makes that run fast and un-sabotageable if someone accepts the gate. Proven real-world results remain the committed SPRTs: NNUE +107.1 Elo versus HCE, incremental accumulators +68.6 Elo, HalfKA v3 -70.3 Elo, data-scale ladder -796 / -383 / -307 Elo versus the shipped net.

VII. REPRODUCTION

`GO_CLOUD=I_ACCEPT_SPRT_GATES DEVICE=cuda BATCH_SIZE=131072 scripts/nnue-pipeline/cloud_train_v4.sh out.bin shard*.bin` Then SPRT with Adaptive=true. Do not promote on val-MAE alone.

REFERENCES

- [1] Stockfish / nnue-pytorch, HalfKAv2_hm feature scheme.
- [2] D. Tan and A. Watkinson Medina, Study of the Proper NNUE Dataset, arXiv:2412.17948.
- [3] Unchessed AI, NNUE v4 full-scale training recipe, docs/nnue-v4-training-recipe.md.
- [4] Unchessed AI, Centipawn validation-loss floors and persona coupling, docs/ieee-low-cp-val-mae-and-persona.md.
- [5] Ruoss et

al., Grandmaster-level chess without search,
arXiv:2402.04494.